Substances in cells I

Major chemical compounds that contribute to cell function

In order for cells to function properly, they require a regular supply of certain substances. Important chemical compounds that contribute to cell function include carbohydrates, lipids, proteins, salts, nucleic acids, water and water-containing substances. A healthy diet ensures that we receive these in the correct ratio.

Carbohydrates

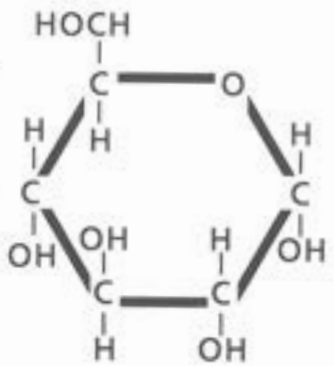
Carbohydrates (meaning *carbon plus water*) are an important source of energy and include sugars, starch and glycogen. They have the general formula $CH_{2x}O_x$, or there are two hydrogen atoms present for every oxygen atom, ie the same ratio as there is in water. Sugars and other carbohydrates react with oxygen to produce energy. Common sugars include glucose, fructose, lactose and sucrose.

1 What do the names of the sugars have in common?

Several types of sugars exist:

- **monosaccharides**: simple sugars such as glucose and fructose (*mono* one).
- **disaccharides**: two sugars such as sucrose and lactose (*di* two). Disaccharides are made of two monosaccharides.
- **polysaccharides**: more complex sugars such as glycogen and starch. Polysaccharides are made of three or more monosaccharides (*poly* many).

Glucose ($C_6H_{12}O_6$) is a normal constituent of the blood and tissue of animals. It consists of a single ring made up of five carbon atoms and one oxygen atom. It is a white, very soluble solid. Glucose (and other, similar sugars) reacts with Benedict's solution to change it from a blue to an orange red colour. Because glucose is so easily assimilated by the body, it is often used in 'energy' drinks and medical 'drips' where fluids are fed directly into veins.



Glucose

2 Is glucose a carbohydrate? Explain your answer.

Sucrose ($C_{12}H_{22}O_{11}$) is obtained in Australia from sugar cane and is soluble in water.

Starch is a complex carbohydrate and has the formula $(C_6H_{10}O_5)x$, where 'x' can be from 50 to many thousands. It is a polysaccharide made up of linked glucose units and is insoluble in water. It is an important sugar storage form for plants. In animals, starch is broken down to simpler sugars in digestion. The test for starch is that it turns iodine solution from yellow/brown to blue/black.

Another polysaccharide molecule is *glycogen* (sometimes called *animal starch*). Glycogen is used to store sugars in animal cells and fungi. In animals it is stored in the liver and muscles. Animals make glycogen from glucose molecules that are not needed straight away. Glycogen is easily broken down to glucose when the organism requires a quick boost of energy.

3 Explain the difference between a monosaccharide, disaccharide and a polysaccharide.

4 Which elements are present in carbohydrates?

5 Draw the structure of a glucose molecule and give its chemical formula.

6 Why is glucose, rather than other sugars, used in 'energy drinks'?

Outline the chemical tests for:

7 glucose

8 starch

9 **Mental challenge!** *Cellulose* is another polysaccharide found in plants. Find out where it is stored in plants and what its main function is.

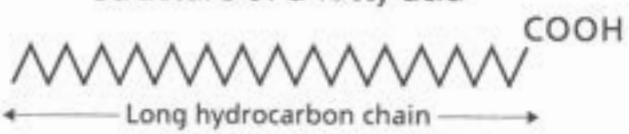
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Lipids

Lipids are fats or fat-like compounds (Gk. *lipos* fat) containing carbon, hydrogen and oxygen. A solid lipid is called a fat and a liquid lipid is an oil. Fats and oils are also sometimes called *triglycerides*. Lipids act as longer term, often permanent, energy storage (as opposed to glycogen, which is a more accessible 'temporary' storage system). They tend to be formed instead of glycogen when the storage space for glycogen is 'full up'. Lipids are not soluble in water. *Phospholipids* are an essential part of the cell membrane and play a part in the flow of water and water-soluble substances in and out of the cell. 'Phospho' denotes phosphorous. Phospholipids are lipids that yield phosphorous.

Fats and oils consist of *fatty acids* (types of organic acids) linked to an alcohol (alkanol), *glycerol*. Typically, fats contain around twice as much energy per gram as proteins and carbohydrates. The test for a fat or oil is that they leave a greasy mark on paper.

Structure of a fatty acid



$CH_3 (CH_{2n}) COOH$ (n = variable number)

Name the elements present in:

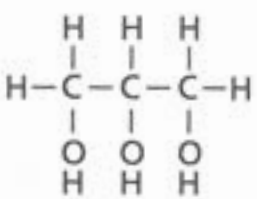
10 a fatty acid

11 glycerol

12 Name a health problem that could result with a diet containing too much fat?

13 What is the major physical difference between a fat and an oil?

Structure of glycerol



Waxes resemble fats and oils but are harder and less greasy. When we think of waxes, beehives and candles often come to mind, but waxes have other important uses. Waxes form coatings on leaves, fruit skins, feathers and fur.

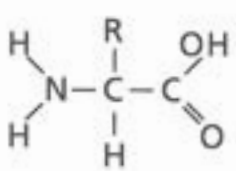
14 Suggest how wax would be useful on feathers, fur and leaves.

Steroids are also present in cells, and resemble lipids in many ways.

Proteins

Proteins (from the Greek, *proteios*, meaning 'primary') are used for cell growth and repair but can also be broken down to provide energy. Proteins are built up from amino acids. There are around 20 different amino acids and these can form many different proteins. There are approximately 30,000 different proteins present in the human body. Enzymes and most hormones are proteins.

Structure of an amino acid



(R = various groups)

15 Give a brief definition for *hormone*.

Plants build up amino acids from substances formed from photosynthesis. These amino acids are converted to proteins. Animals take in proteins in their diet and break these down (digest) them into amino acid components. These amino acids are carried by the bloodstream to cells, where they are reassembled into the specific proteins required by these cells.

'Biuret' reagent can be used to test for proteins. This solution contains copper ions in alkaline solution and goes purple when proteins are present. The greater the amount of protein the deeper the purple colour.

16 What is the simple test for lipids?

17 Name one element present in amino acids but not fatty acids.

18 Are lipids soluble in water?

19 Name two uses cells make of proteins.

20 Outline how you could test if a substance contained proteins.

21 True or false? Proteins contain twice as much energy per gram as fats and oils.

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